# Asymptotic Notations

# Mathematical Model of Computation

## Framework for Algorithmic Analysis

## Need of Mathematical Model

Real Computers Vs Mathematical Model

# Mathematical Model of Computation

Model of Computation specifies:

- What operations in an algorithm is allowed
- Cost (time) of each operation

Two components of Model

- Processor (has defined instruction set)
- Memory (collection of locations)

# Computing running time in Model

Simple Statement

(1 step execution )

- Arithmetic & logical operations.
- Jumps & conditional jumps
- Pointer instruction

- Array operations (1D) [2 step execution]

- More Complex Statement

(multistep execution)

A=B+C*D-F 3 steps

A[I]=B[I] + C[I] 4 steps

X=B[I]
Y=C[I]
Z=X+Y
A[I]=Z

C = A[I,J]4 steps

A' [(I)*M + J] where m is the number of columns

# General Analysis Strategy T(n)

T(n)= n

- Max time taken by an algorithm to solve any instance of size n.
- measures a goodness of an algorithm, i.e functional form of T(n) is important (linear, cubic, quadratic, etc.)
- Significance of T(n) is independent of model/computer we used for analysis because here the objective is to characterize the algorithm not the machine

**Running Time Calculations**

To Add elements of two arrays of size n

# Asymptotic Notations

Has two features

1.Function classification

2.Identify the behaviour as n→

# Asymptotic Complexity

- Running time of an algorithm as a function of input size $n$ **for large $n$**.

- Expressed using only the **highest-order term** in the expression for the exact running time.
  - Instead of exact running time, say $\Theta(n^2)$.

- Written using ***Asymptotic Notation***.

# Asymptotic Notation

- $\Omega$, $O$, $\Theta$, $o$, $\omega$

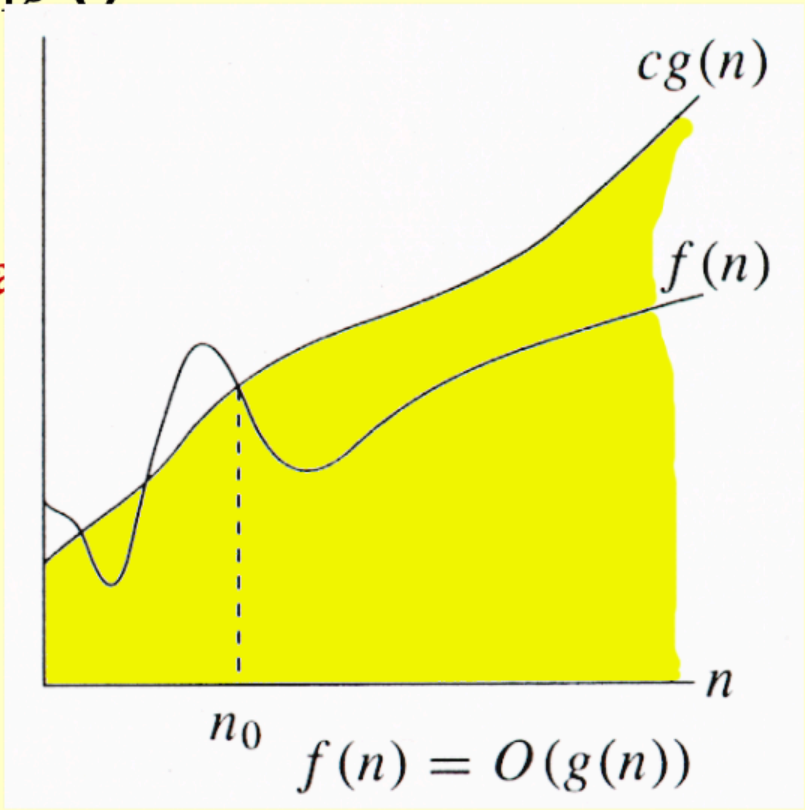
- Defined for functions over the natural numbers.
  - Ex: $f(n) = \Theta(n^2)$.
  - Describes how $f(n)$ grows in comparison to $n^2$.

- The notations describe different rate-of-growth relations between the defining function and the defined set of functions.

# *O*-notation

For function $g(n)$, we define $O(g(n))$, big-O of $n$, as the set:

$O(g(n)) = \{f(n) :$
$\exists$ positive constants $c$ and $n_0$ such that
$\forall n \geq n_0$,
we have $0 \leq f(n) \leq cg(n)\}$




***Intuitively***: Set of all functions whose *rate of growth* is the same as or lower than that of $g(n)$.

**$g(n)$ is an *asymptotic upper bound* for $f(n)$.**

$f(n) = \Theta(g(n)) \Rightarrow f(n) = O(g(n))$.
$\Theta(g(n)) \subseteq O(g(n))$.

# Examples

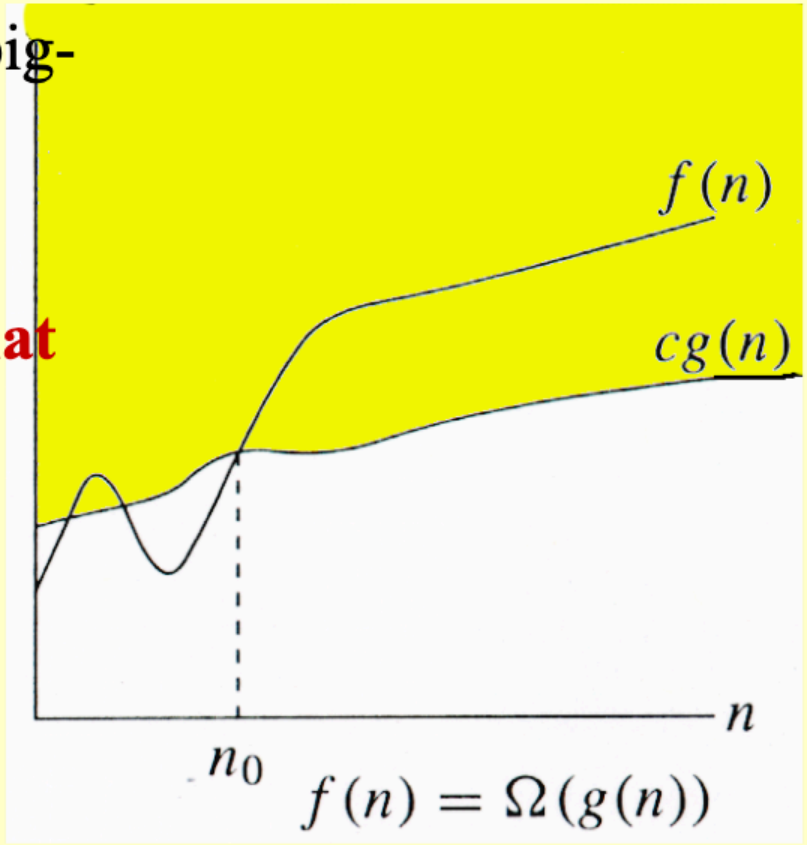
$O(g(n)) = \{f(n)$ : $\exists$ positive constants $c$ and $n_0$, such that $\forall n \geq n_0$, we have $0 \leq f(n) \leq cg(n)\}$

- Any linear *function* $an + b$ is in $O(n^2)$. **How?**
- Show that $3n^3 = O(n^4)$ for appropriate $c$ and $n_0$.

# □ -notation

For function $g(n)$, we define □($g(n)$), big-Omega of $n$, as the set:

□($g(n)$) = {$f(n)$ :
□ positive constants $c$ and $n_0$, such that
□$n$ □ $n_0$,
we have 0 □ $cg(n)$ □ $f(n)$}




***Intuitively***: Set of all functions whose *rate of growth* is the same as or higher than that of $g(n)$.

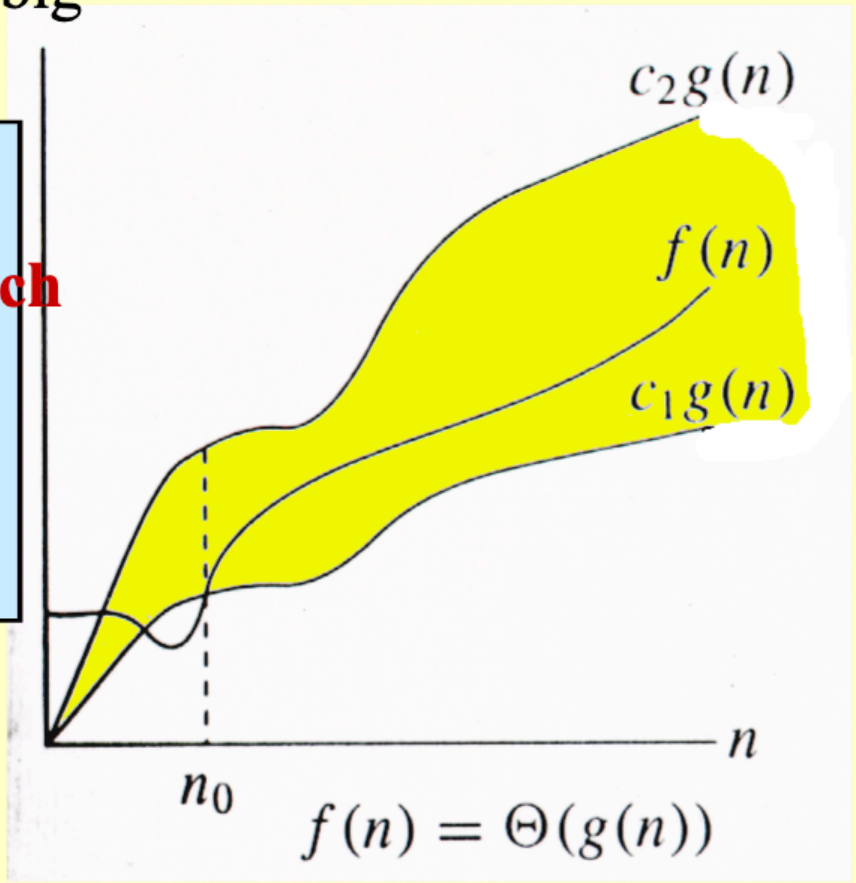
**$g(n)$ is an *asymptotic lower bound* for $f(n)$.**

$f(n)$ = □($g(n)$) □ $f(n)$ = □($g(n)$).
□($g(n)$) □ □($g(n)$).

# □-notation

For function $g(n)$, we define □($g(n)$), big-Theta of $n$, as the set:

□($g(n)$) = {$f(n)$ :
□ positive constants $c_1$, $c_2$, and $n_0$, such that □$n$ □ $n_0$,
we have 0 □ $c_1 g(n)$ □ $f(n)$ □ $c_2 g(n)$
}



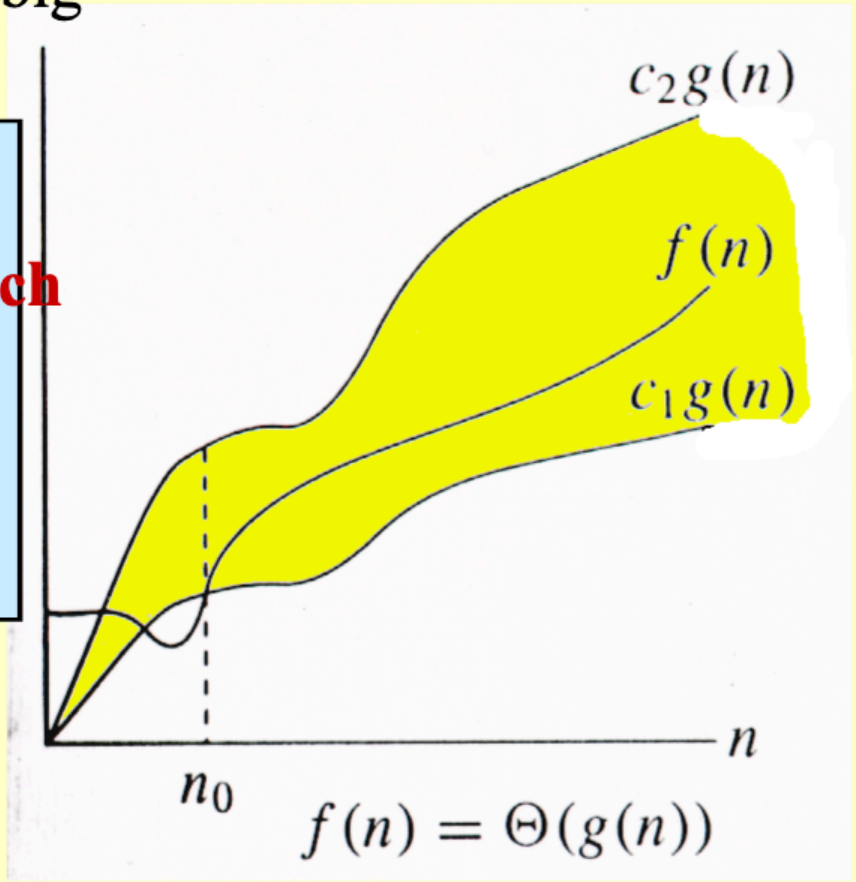

***Intuitively***: Set of all functions that have the same *rate of growth* as $g(n)$.

**$g(n)$ is an *asymptotically tight bound* for $f(n)$.**

# □-notation

For function $g(n)$, we define □($g(n)$), big-Theta of $n$, as the set:

□($g(n)$) = {$f(n)$ :
□ positive constants $c_1$, $c_2$, and $n_0$, such that □$n$ □ $n_0$,
we have 0 □ $c_1 g(n)$ □ $f(n)$ □ $c_2 g(n)$
}




Technically, $f(n)$ □ □($g(n)$).
Older usage, $f(n)$ = □($g(n)$).
accept either notation

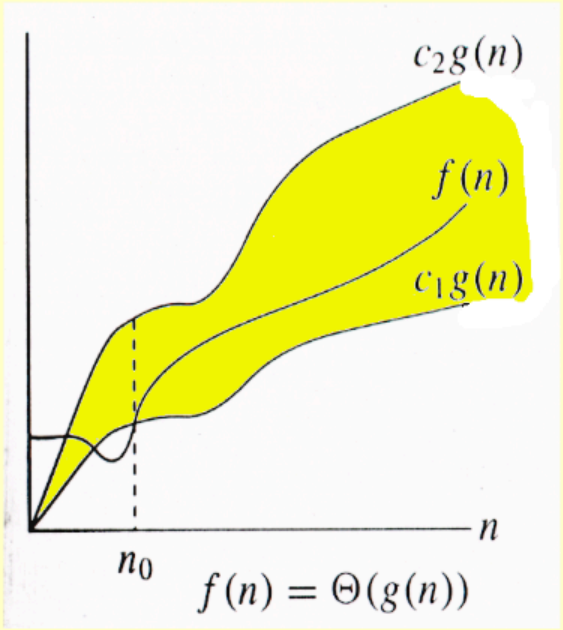
**$f(n)$ and $g(n)$ are nonnegative, for large $n$.**

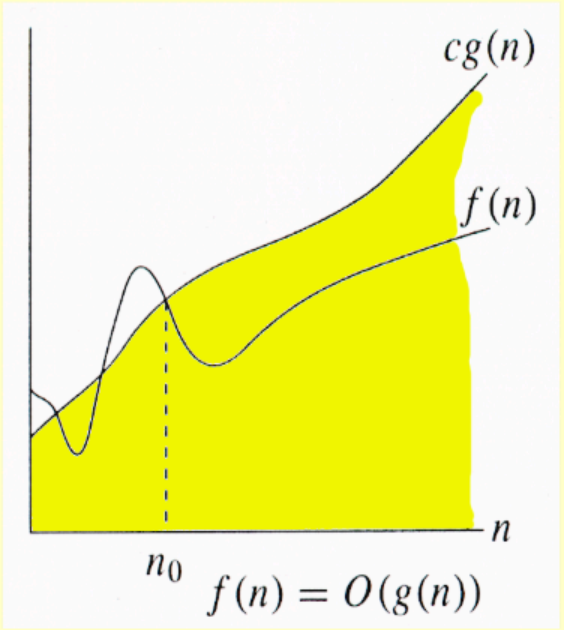
# Example

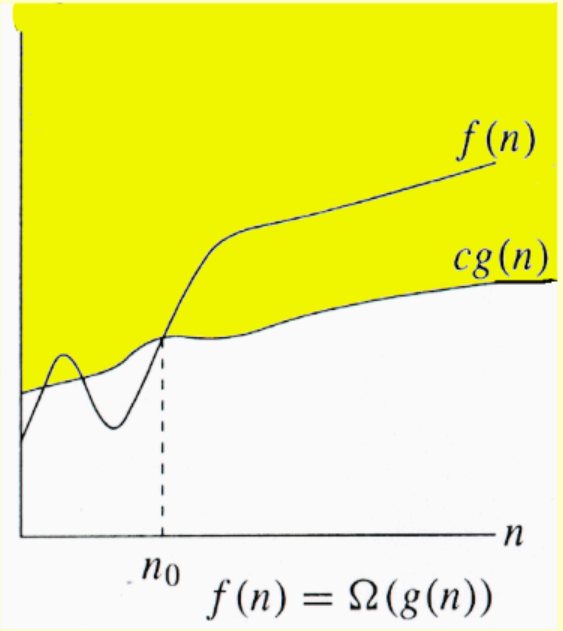
$\Theta(g(n)) = \{f(n) :$ positive constants $c_1$, $c_2$, and $n_0$, such that $n \geq n_0$, $0 \leq c_1 g(n) \leq f(n) \leq c_2 g(n)\}$

- $10n^2 - 3n = \Theta(n^2)$
- What constants for $n_0$, $c_1$, and $c_2$ will work?
- Make $c_1$ a little smaller than the leading coefficient, and $c_2$ a little bigger.
- ***To compare orders of growth, look at the leading term.***

# Relations Between $\Theta$, $O$, $\Omega$



# Relations Between □, □, *O*

**Theorem :** For any two functions $g(n)$ and $f(n)$,

$f(n) = \square(g(n))$ iff

$f(n) = O(g(n))$ **and** $f(n) = \square(g(n))$.

- I.e., $\square(g(n)) = O(g(n)) \cap \square(g(n))$
- In practice, asymptotically tight bounds are obtained from asymptotic upper and lower bounds.

# Running Times

- "Running time is $O(f(n))$" □ Worst case is $O(f(n))$
- "Running time is □$(f(n))$" □ Best case is □$(f(n))$

# Comparison of Functions

$$f \square g \square a \square b$$

$$f(n) = O(g(n)) \square a \square b$$

$$f(n) = \square(g(n)) \square a \square b$$

$$f(n) = \square(g(n)) \square a = b$$

$$f(n) = o(g(n)) \square a < b$$

$$f(n) = \square(g(n)) \square a > b$$

# Properties

- **Transitivity**

$$f(n) = \Theta(g(n)) \;\&\; g(n) = \Theta(h(n)) \Rightarrow f(n) = \Theta(h(n))$$
$$f(n) = O(g(n)) \;\&\; g(n) = O(h(n)) \Rightarrow f(n) = O(h(n))$$
$$f(n) = \Omega(g(n)) \;\&\; g(n) = \Omega(h(n)) \Rightarrow f(n) = \Omega(h(n))$$
$$f(n) = o(g(n)) \;\&\; g(n) = o(h(n)) \Rightarrow f(n) = o(h(n))$$
$$f(n) = \omega(g(n)) \;\&\; g(n) = \omega(h(n)) \Rightarrow f(n) = \omega(h(n))$$

- **Reflexivity**

$$f(n) = \Theta(f(n))$$
$$f(n) = O(f(n))$$
$$f(n) = \Omega(f(n))$$

# Properties

- **Symmetry**

$$f(n) = \Omega(g(n)) \text{ iff } g(n) = \Omega(f(n))$$

- **Complementarity**

$$f(n) = O(g(n)) \text{ iff } g(n) = \Omega(f(n))$$

$$f(n) = o(g(n)) \text{ iff } g(n) = \omega((f(n))$$

# Growth of a function

Find the **big oh** for the following functions

- $f(n) = 3n + 5$
- $f(n) = 27n^2 + 16n + 25$
- $f(n) = 3n^3 + 4n$
- $f(n) = 2^n + 6n^2 + 3n$
- $f(n) = 5*2^n + 3n + 5$
- $f(n) = 3*2^n + 4n^2 + 5n + 3$

# Incorrect bounds

- $7n+5 \neq O(1)$
- $27n^2 + 16n + 25 \neq O(n)$
- $5n^3 + n^2 + 3n + 2 \neq O(n^2)$

# Find the **big Omega** for the following functions

- $f(n) = 27n^2 + 16n$
- $f(n) = 4n^3 + 2n + 3$
- $f(n) = 2^n + 6n^2 + 3n$
- $f(n) = 5*2^n + 3n + 5$

# Incorrect bounds

- $7n+5 \neq \square(n^2)$
- $5n^2 + 4n + 2 \neq \square(n^3)$
- $3n^3 + 5n^2 + 2n \neq \square(n^4)$

# Find the **Theta** for the following functions

- $f(n) = 27n^2 + 16n$
- $f(n) = 4n^3 + 2n + 3$
- $f(n) = 2^n + 6n^2 + 3n$

# Incorrect bounds

- $7n+5 \neq \square(1)$
- $10n^2+7 \neq \square(n^3)$
- $5n+8 \neq \square(n^2)$

# *Related Contents*

## *Basics of Asymptotic Notations*

https://cexpertvision.com/2023/06/23/basics-of-asymptotic-notations/

Video Lecture of the topic can be found at

**Mathematical Model of Computation**

**https://youtu.be/JEnXJPlXhng**

**Basics of Asymptotic Notations**

**https://youtu.be/OQZySiRblPo**